

HANKO TVARMINNE, FinlandWMO: **027500**Lat: **59.8438N**Lon: **23.2485E**Elev: **10**StdP: **14.69**Time Zone: **2.00 (EUE)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.6	5.8	-3.3	4.6	1.2	1.5	5.9	7.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	76.1	68.2	73.5	66.2	70.8	64.2	69.6	74.9	67.6	72.3	65.4	69.6	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.5	101.1	73.1	65.6	94.6	70.9	63.5	87.9	68.5	33.6	74.9	32.0	72.3	30.3	69.8	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-0.9	79.5	8.0	4.3	-6.7	82.6	-11.4	85.1	-15.9	87.5	-21.7	90.6		
(5) N/A	N/A	N/A	-1.4	70.8	7.9	2.8	-7.1	72.8	-11.8	74.4	-16.2	76.0	-22.0	78.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.9	25.5	26.1	30.9	38.8	49.6	57.3	64.0	62.4	55.4	44.6	38.7	32.3
	DBStd	14.75	8.92	9.81	6.82	4.95	6.76	4.60	4.99	4.05	4.68	5.70	6.57	7.92
	HDD50	3532	759	668	594	338	91	2	0	0	13	180	338	549
	HDD65	7773	1224	1088	1059	786	476	234	82	103	287	632	788	1014
	CDD50	1309	0	0	0	2	80	222	433	383	176	13	0	0
	CDD65	76	0	0	0	0	1	3	51	21	0	0	0	0
	CDH74	192	0	0	0	0	0	7	154	31	0	0	0	0
	CDH80	8	0	0	0	0	0	0	7	1	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	26.7	2.4	1.7	1.5	1.4	1.3	2.0	2.4	3.3	2.2	2.9	3.1	2.6
	PrecMax	35.9	4.2	4.1	3.6	2.4	2.8	4.3	6.8	7.2	5.2	7.3	7.1	6.1
	PrecMin	19.5	0.3	0.1	0.3	0.2	0.2	0.5	0.2	0.3	0.7	0.8	0.3	0.8
	PrecStd	3.8	1.0	1.0	0.7	0.6	0.7	0.9	1.5	1.7	1.1	1.5	1.4	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.5	41.5	46.9	59.0	71.8	73.8	80.8	77.4	67.0	57.6	50.7	47.1
		MCWB	39.2	36.9	40.8	47.5	58.6	62.0	68.1	69.6	61.9	54.1	49.3	45.1
	2%	DB	39.1	38.2	43.2	53.6	66.5	70.2	78.0	74.1	65.0	55.5	49.2	45.2
		MCWB	37.9	36.2	38.0	45.8	56.1	60.9	68.9	67.2	60.7	53.3	48.1	43.5
	5%	DB	37.8	36.9	40.4	49.5	63.6	67.4	75.4	71.4	63.6	54.1	47.7	43.5
		MCWB	36.8	35.6	36.6	42.5	55.4	60.0	68.0	65.3	59.8	52.2	46.5	41.9
	10%	DB	36.2	35.8	38.4	46.2	60.6	65.4	72.9	69.3	62.0	52.7	46.4	41.8
		MCWB	35.3	34.9	35.6	41.1	54.0	58.5	66.3	63.8	58.8	50.7	45.1	40.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.7	38.0	41.6	48.4	60.5	65.7	71.7	71.0	63.4	55.3	50.0	45.9
		MCDB	40.1	39.8	46.4	56.4	67.0	69.4	76.6	76.1	65.6	56.2	50.5	46.7
	2%	WB	38.4	36.8	38.8	46.1	58.3	63.0	70.4	68.4	61.9	54.1	48.4	43.9
		MCDB	39.0	37.7	41.7	53.3	64.1	68.1	75.8	72.7	64.2	55.0	48.9	44.8
	5%	WB	37.0	35.9	37.5	43.9	56.4	61.2	69.0	66.6	60.6	52.9	46.8	42.1
		MCDB	37.7	36.7	39.6	48.5	62.0	65.9	74.4	70.5	62.7	53.9	47.6	43.2
	10%	WB	35.5	34.9	36.2	41.9	54.3	59.5	67.0	64.7	59.4	51.1	45.3	40.3
		MCDB	36.1	35.7	37.8	45.3	60.0	64.1	71.9	68.1	61.5	52.3	46.2	41.5
Mean Daily Temperature Range	5% DB	MDBR	8.1	7.7	9.4	10.3	11.5	11.2	11.1	10.1	8.6	7.6	5.6	6.8
		MCDBR	5.8	6.5	10.4	15.4	15.1	14.3	13.3	12.1	10.1	8.0	6.0	7.1
		MCWBR	5.6	5.6	6.9	9.5	8.3	7.8	7.0	6.8	6.8	6.9	6.0	7.0
	5% WB	MCDBR	6.0	6.1	8.8	14.0	13.2	12.6	11.8	10.2	8.6	7.1	5.7	6.9
		MCWBR	6.0	5.5	6.6	8.9	8.3	7.4	6.9	6.8	6.4	6.8	6.0	7.0
Clear-Sky Solar Irradiance	taub	0.280	0.313	0.334	0.373	0.367	0.363	0.390	0.383	0.352	0.335	0.311	0.266	
	taud	2.158	2.221	2.288	2.317	2.415	2.467	2.415	2.424	2.495	2.404	2.219	2.093	
	Ebn at Noon	155	215	252	262	273	276	264	257	247	209	141	111	
	Edh at Noon	14	23	29	34	33	32	33	30	24	19	13	10	
All-Sky Solar Radiation	RadAvg	77	244	727	1255	1755	1853	1748	1377	838	364	99	49	
	RadStd	12	44	86	131	171	120	149	136	103	50	19	6	